

10. Joint distributions

I. Sum of independent random variables

Let X and Y be independent random variables on the same probability space. How can we describe the distribution of $X+Y$?

Prop. Let X & Y be independent integer valued random variables, then for any integer $k \in \mathbb{Z}$

$$P(X+Y=k) = \sum_{i=-\infty}^{\infty} P(X=k-i)P(Y=i) = \sum_{i=-\infty}^{\infty} P(X=i)P(Y=k-i)$$

proof:

$$\underline{P(X+Y=k)} = \sum_{i=-\infty}^{\infty} P(\{X+Y=k\} \cap \{Y=i\}) = \sum_{i=-\infty}^{\infty} P(\{X=k-i\} \cap \{Y=i\}) = \sum_{i=-\infty}^{\infty} P(X=k-i)P(Y=i) \quad \square$$

Prop. Let $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$ be independent random variables. Then $X+Y \sim \text{Poi}(\lambda+\mu)$.

proof: For $k \geq 0$

$$\underline{P(X+Y=k)} = \sum_{i=-\infty}^{\infty} P(X=k-i)P(Y=i) = \sum_{i=0}^k e^{-\lambda} \frac{\lambda^{k-i}}{(k-i)!} e^{-\mu} \frac{\mu^i}{i!} = e^{-(\lambda+\mu)} \frac{1}{k!} \sum_{i=0}^k \binom{k}{i} \lambda^{k-i} \mu^i = e^{-(\lambda+\mu)} \frac{(\lambda+\mu)^k}{k!} \quad \square$$

Prop. If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ are independent, then $X+Y \sim \text{Bin}(n+m, p)$

$$\begin{aligned} \underline{\text{proof:}} \quad P(X+Y=k) &= \sum_{i=0}^k P(X=k-i)P(Y=i) = \sum_{i=0}^k \binom{n}{k-i} p^{k-i} (1-p)^{n-k+i} \binom{m}{i} p^i (1-p)^{m-i} \\ &= p^k (1-p)^{n+m-k} \sum_{i=0}^k \binom{n}{k-i} \binom{m}{i} \stackrel{\text{Ex. 1.17.}}{=} \binom{n+m}{k} p^k (1-p)^{n+m-k} \quad \square \end{aligned}$$

Prop. Let X, Y be independent, absolutely continuous random variables.

Let F_X, F_Y be the distribution functions, f_X, f_Y be the density functions of X and Y respectively. Then

$$F_{X+Y}(a) = \int_{-\infty}^{\infty} F_X(a-y) f_Y(y) dy = \int_{-\infty}^{\infty} F_Y(a-x) f_X(x) dx$$

proof: Denote the joint density of X, Y with $f(x, y)$, then $f(x, y) = f_X(x) f_Y(y)$.

$$F_{X+Y}(a) = P(X+Y < a) = \iint_{\{x+y < a\}} f(x, y) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{a-y} f_X(x) f_Y(y) dx dy = \int_{-\infty}^{\infty} F_X(a-y) f_Y(y) dy \quad \square$$

Cor. 1 If X, Y are independent, absolutely continuous random variables, then the density function of $X+Y$ is

$$f_{X+Y}(a) = \int_{-\infty}^{\infty} f_X(a-y) f_Y(y) dy$$

Prop. If $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ are independent, then

$$X+Y \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$$

proof: Let $\sigma_1^2 = \sigma_1^2/\sigma_1^2$ and define the independent random variables

$$X' \sim N(0, 1), \quad Y' \sim N(0, \sigma^2)$$

Then, it is enough to prove $X'+Y' \sim N(0, 1+\sigma^2)$, since

$$\left. \begin{aligned} X &\sim N(\mu_1, \sigma_1^2) \Rightarrow X = \sigma_1 X' + \mu_1 \\ Y &\sim N(\mu_2, \sigma_2^2) \Rightarrow Y = \sigma_1 Y' + \mu_2 \end{aligned} \right\} X+Y = \sigma_1(X'+Y') + \mu_1 + \mu_2 \sim N(\mu_1+\mu_2, \sigma_1^2 + \sigma_2^2)$$

$$\begin{aligned} f_{X'}(a-y) f_{Y'}(y) &= \frac{1}{\sqrt{2\pi}} e^{-\frac{(a-y)^2}{2}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} = \frac{1}{2\pi\sigma} e^{-\frac{y^2 + \sigma^2 y^2 - 2a\sigma^2 y + a^2\sigma^2}{2\sigma^2}} = \\ &= \frac{1}{2\pi\sigma} e^{-\frac{1+\sigma^2}{2\sigma^2} \left(y^2 - 2a\frac{\sigma^2}{1+\sigma^2} y + a^2\frac{\sigma^2}{1+\sigma^2} \right)} = \frac{1}{2\pi\sigma} e^{-\frac{1+\sigma^2}{2\sigma^2} \left(\left(y - a\frac{\sigma^2}{1+\sigma^2} \right)^2 - a^2\frac{\sigma^4}{(1+\sigma^2)^2} + a^2\frac{\sigma^2}{1+\sigma^2} \right)} = \\ &= \frac{1}{2\pi\sigma} e^{-\frac{1+\sigma^2}{2\sigma^2} \left(y - a\frac{\sigma^2}{1+\sigma^2} \right)^2 - \frac{a^2}{2} \left(-\frac{\sigma^2}{1+\sigma^2} + 1 \right)} = \frac{1}{2\pi\sigma} e^{-\frac{1+\sigma^2}{2\sigma^2} \left(y - a\frac{\sigma^2}{1+\sigma^2} \right)^2} e^{-\frac{a^2}{2(1+\sigma^2)}} \end{aligned}$$

$$\begin{aligned} \Rightarrow f_{X'+Y'}(a) &= \int_{-\infty}^{\infty} f_{X'}(a-y) f_{Y'}(y) dy = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma} e^{-\frac{1+\sigma^2}{2\sigma^2} \left(y - a\frac{\sigma^2}{1+\sigma^2} \right)^2} \cdot e^{-\frac{a^2}{2(1+\sigma^2)}} dy = \\ &= \frac{1}{\sqrt{2\pi}\sqrt{1+\sigma^2}} e^{-\frac{a^2}{2(1+\sigma^2)}} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\frac{\sigma}{\sqrt{1+\sigma^2}}} e^{-\frac{\left(y - a\frac{\sigma^2}{1+\sigma^2} \right)^2}{2\frac{\sigma^2}{1+\sigma^2}}} dy}_{=1, \text{ density of } N\left(a\frac{\sigma^2}{1+\sigma^2}, \frac{\sigma^2}{1+\sigma^2}\right)} = \frac{1}{\sqrt{2\pi}(1+\sigma^2)} e^{-\frac{a^2}{2(1+\sigma^2)}} \end{aligned}$$

$$\Rightarrow X'+Y' \sim N(0, 1+\sigma^2)$$

Example, A basketball team plays $n_A=26$ matches in league A and $n_B=18$ matches in league B. They win with probability $p_A=0.4$ and $p_B=0.7$ in league A, B respectively. Suppose that the outcome of the matches are independent. □

$X_A = \{\text{matches they win in league A}\}$ $X_B = \{\text{matches they win in league B}\}$
What is the probability, that all together they win more than 25 matches?

$X_A \sim \text{Bin}(n_A, p_A)$ and $X_B \sim \text{Bin}(n_B, p_B)$, but unfortunately $p_A \neq p_B$ so $X_A + X_B$ is not a binomial distribution.

↳ Let's apply the DeMoivre-Laplace theorem!
Since the sum of independent normal distributions ^{is} always normal as well.

$$\frac{X_A - \eta_A p_A}{\sqrt{\eta_A p_A (1-p_A)}} \sim N(0,1), \quad \frac{X_B - \eta_B p_B}{\sqrt{\eta_B p_B (1-p_B)}} \sim N(0,1)$$

$$P(X_A + X_B \geq 25) = P(X_A + X_B \geq 24.5) = \leftarrow c \in \mathbb{R} \text{ is a constant, we define it later}$$

$$= P\left(\underbrace{\frac{X_A - \eta_A p_A}{\sqrt{\eta_A p_A (1-p_A)}}}_{\sim N(0,1)} \underbrace{\frac{\sqrt{\eta_A p_A (1-p_A)}}{c}}_c + \underbrace{\frac{X_B - \eta_B p_B}{\sqrt{\eta_B p_B (1-p_B)}}}_{\sim N(0,1)} \underbrace{\frac{\sqrt{\eta_B p_B (1-p_B)}}{c}}_c \geq \frac{24.5 - \eta_A p_A - \eta_B p_B}{c} \right) =$$

$$\underbrace{\sim N\left(0, \frac{\eta_A p_A (1-p_A)}{c^2}\right)}_{\sim N\left(0, \frac{\eta_A p_A (1-p_A) + \eta_B p_B (1-p_B)}{c^2}\right)} \underbrace{\sim N\left(0, \frac{\eta_B p_B (1-p_B)}{c^2}\right)}_{\sim N\left(0, \frac{\eta_A p_A (1-p_A) + \eta_B p_B (1-p_B)}{c^2}\right)}$$

To obtain a standard normal distribution we choose

$$c = \sqrt{\eta_A p_A (1-p_A) + \eta_B p_B (1-p_B)}$$

$$= 1 - \Phi\left(\frac{24.5 - \eta_A p_A - \eta_B p_B}{\sqrt{\eta_A p_A (1-p_A) + \eta_B p_B (1-p_B)}}\right) \approx 1 - \Phi(0.474) \approx \underline{\underline{0.32}}$$

II. Conditional distributions

def, Let X and Y be discrete random variables, and choose a y for which $P(Y=y) > 0$. Then the conditional weight function of X conditioned to $Y=y$ is

$$P_{X|Y}(x|y) := P(X=x|Y=y) = \frac{P(\{X=x\} \cap \{Y=y\})}{P(Y=y)}$$

Example: The joint weight function of X and Y is given by this table

$X \backslash Y$	0	1
0	0.4	0.2
1	0.1	0.3

What is the conditional distribution of X given $\{Y=0\}$?

$$P_{X|Y}(0|0) = \frac{P(0,0)}{P(0,0) + P(1,0)} = \frac{0.4}{0.5} = \underline{\underline{\frac{4}{5}}}$$

$$P_{X|Y}(1|0) = \frac{P(1,0)}{P(0,0) + P(1,0)} = \frac{0.1}{0.5} = \underline{\underline{\frac{1}{5}}}$$

where $p(x,y) = P(X=x, Y=y)$ is the joint weight function.

def, Let X, Y be jointly absolutely continuous random variables, and let y be such that $f_Y(y) > 0$. Then the conditional distribution function of X conditioned to $Y=y$ is

$$F_{X|Y}(x|y) := \frac{\int_{-\infty}^x f_{XY}(x,y) dx}{f_Y(y)},$$

and the conditional density is

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)}$$

($F(x,y)$ = joint distribution

$f(x,y)$ = joint density

$f_Y(y)$ = marginal density of Y)

Note: Since $P(Y=y)=0$, $\forall y$ in the continuous case,

" $F_{X|Y}(x|y) := P(X \leq x | Y=y)$ " is not a usable definition. Instead,

$$F_{X|Y}(x|y) := \lim_{\varepsilon \rightarrow 0^+} P(X \leq x | Y \in [y, y+\varepsilon)) = \lim_{\varepsilon \rightarrow 0^+} \frac{P(X \leq x, Y \in [y, y+\varepsilon))}{P(Y \in [y, y+\varepsilon))} =$$

$$= \lim_{\varepsilon \rightarrow 0^+} \frac{F(x, y+\varepsilon) - F(x, y)}{F_Y(y+\varepsilon) - F_Y(y)} \frac{\varepsilon}{\varepsilon} = \frac{\frac{\partial}{\partial y} F(x, y)}{f_Y(y)}$$

Example: Let $X \sim \text{Uni}[0,1]$ and $Y|X=x \sim \text{Uni}[0, x]$ if $0 < x < 1$.

$f_Y(y) = ?$ $f_{X|Y}(x|y) = ?$

$$f_X(x) = \begin{cases} 1, & x \in [0,1] \\ 0, & \text{otherwise} \end{cases} \quad f_{Y|X}(y|x) = \begin{cases} \frac{1}{x}, & y \in [0, x] \text{ and } x \in (0,1) \\ 0, & \text{otherwise} \end{cases}$$

$$f(x, y) = f_{Y|X}(y|x) f_X(x) = \begin{cases} \frac{1}{x}, & 0 < y \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\Rightarrow \underline{f_Y(y)} = \int_{-\infty}^{\infty} f(x, y) dx = \int_y^1 \frac{1}{x} dx = -\underline{\ln y} \quad \text{for } y \in (0,1)$$

$$\underline{f_{X|Y}(x|y)} = \frac{f(x, y)}{f_Y(y)} = \begin{cases} -\frac{1}{x \ln y}, & 0 < y \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$$

III.] Properties of the expected value]

Recall: X is discrete $\Rightarrow E[X] = \sum_i x_i \cdot P(X=x_i)$

X is absolutely continuous $\Rightarrow E[X] = \int x f(x) dx$

$\forall a, b \in \mathbb{R}, E[aX+b] = aE[X] + b$

Prop: Let $a < b$ be real numbers such that $P(a \leq X \leq b) = 1$. Then $a \leq E[X] \leq b$.

Lemma: Let X be a non-negative random variable, that is $P(X \geq 0) = 1$.
Then $E[X] = \int_0^{\infty} P(X > t) dt$

proof: We've already showed it for integer valued random variables in Ex. 4.11.
Now, we only prove it for an absolutely continuous X .

$$\int_0^{\infty} P(X > t) dt = \int_0^{\infty} \int_t^{\infty} f(x) dx dt = \int_0^{\infty} \int_0^x f(x) dt dx = \int_0^{\infty} x f(x) dx = E[X]. \quad \square$$

change the order of the integrals

Prop. Let X and Y be random variables, and let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$.

Then $E[g(X, Y)] = \begin{cases} \sum_{x,y} g(x,y) P(X=x, Y=y) & , \text{ if they are jointly discrete} \\ \iint g(x,y) f(x,y) dx dy & , \text{ if they are jointly absolutely continuous} \end{cases}$

proof. We only prove the absolutely continuous case, the discrete one is analogous.

First, suppose that $g: \mathbb{R}^2 \rightarrow \mathbb{R}_+$. Then

$$E[g(X, Y)] \stackrel{\text{lemma}}{=} \int_0^\infty P(g(X, Y) > t) dt = \int_0^\infty \iint_{\substack{f(x,y) > 0 \\ g(x,y) > t}} f(x,y) dx dy dt \quad \leftarrow \text{change the order of the integrals}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_0^{g(x,y)} f(x,y) dt dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f(x,y) dx dy$$

Now let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be arbitrary, and define

$$g_+(x,y) = \max\{0, g(x,y)\}, \quad g_-(x,y) = \max\{0, -g(x,y)\}$$

$$E[g(X, Y)] \stackrel{g = g_+ - g_-}{=} E[g_+(X, Y)] - E[g_-(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{(g_+(x,y) - g_-(x,y))}_{= g(x,y)} f(x,y) dx dy \quad \square$$

Cor. 1 For any X and Y random variables $E[X \pm Y] = E[X] \pm E[Y]$.

Prop. Let X, Y be random variables with $P(X \leq Y) = 1$. Then $E[X] \leq E[Y]$.

proof.

Let $Z = Y - X$, then $1 = P(X \leq Y) = P(Y - X \geq 0) = P(Z \geq 0)$

$$\Rightarrow 0 \leq E[Z] = E[Y - X] \stackrel{\substack{\uparrow \\ \text{previous corollary}}}{=} E[Y] - E[X] \quad \square$$

Prop. (Jensen's inequality)

Let X be a random variable and g be convex, then $E[g(X)] \geq g(E[X])$

eg. For $g(x) = x^2$ we've seen $E[X^2] \geq E[X]^2$

proof. $\forall x_0 \in \mathbb{R}, g(X) \geq g(x_0) + g'(x_0)(X - x_0) \leftarrow g(X) \text{ is above the tangent line at } x_0 \text{ since } g \text{ is convex.}$

$$\Rightarrow E[g(X)] \geq \underbrace{E[g(x_0)]}_{= g(x_0)} + g'(x_0) \underbrace{E[X - x_0]}_{= E[X] - x_0}$$

By choosing $x_0 = E[X]$

$$E[g(X)] \geq g(E[X]) + g'(E[X])(\underbrace{E[X] - E[X]}_{=0}) = \underline{g(E[X])} \quad \square$$

Example: Let $X_1, \dots, X_n$ be identically distributed random variables.

$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the mean value of the random variables.

If $\mathbb{E}[X_i] = \mu$, then $\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu$.

Example: Let $A_1, \dots, A_n$ be events, and let $X_i = \mathbb{1}_{A_i}$.

$$X = \sum_{i=1}^n X_i, \quad Y = \mathbb{1}_{\bigcup_{i=1}^n A_i} \Rightarrow Y \leq X$$

Hence, $\mathbb{P}(\bigcup_{i=1}^n A_i) = \mathbb{E}[Y] \leq \mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n \mathbb{P}(A_i)$

$\hookrightarrow$ It is an alternative proof for $\mathbb{P}(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n \mathbb{P}(A_i)$.

Example: (Binomial distribution)

Suppose that we make n independent trials, where the probability of success is p .

$$X_i = \begin{cases} 1, & \text{ith trial succeeded} \\ 0, & \text{ith trial failed} \end{cases}$$

$$X = \{\text{total number of successful trials}\} =$$

$$= \sum_{i=1}^n X_i \sim \text{Bin}(n, p)$$

$$\underline{\underline{\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = np}}$$

Example: Let $X_1, \dots, X_n$ be independent random variables with distribution $\text{Geo}(p)$.

Let $Y \sim \text{Neg}(n, p)$ be defined as $Y = X_1 + \dots + X_n$, then

$$\underline{\underline{\mathbb{E}[Y] = \sum_{i=1}^n \mathbb{E}[X_i] = \frac{n}{p}}}$$

Example: (Hypergeometric distribution)

There are N deer on a field, and m -many of them are signed.
We collect n deer.

X = number of signed deer we collected

$$Y_i = \begin{cases} 1, & \text{ith deer is signed} \\ 0, & \text{ith deer is not signed} \end{cases} \Rightarrow X = \sum_{i=1}^n Y_i$$

$$\underline{\underline{\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[Y_i] = \sum_{i=1}^n \mathbb{P}(\text{ith deer is signed}) = \sum_{i=1}^n \frac{m}{N} = \frac{m \cdot n}{N}}}$$